

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHANGA

M.Sc. (Nano Science & Technology) examination December 2009
NT702 Concept of Chemical Sciences

Date: 2nd Dec 2009

Time: 10.00 to 10.30

Total marks: 20

PART-A (Do as directed) (Write the answers within the space provided)

1. A mass of oxygen occupies 7 L under pressure of 740 Torr. Determine the volume of the mass of a gas at std. pressure, the temperature remaining constant. Show the calculation and tick the correct one. (3)

(i) 8.62 L (ii) 7.62 L (iii) 6.82 L (iv) 5.82 L

2. Product of pressure time's volume has dimension of energy (show the calculation) (2)

3. A gas at a pressure of 5.0 atm is heated from 0⁰ to 546 ⁰C and simultaneously compressed to one-third of its original volume. What will be the pressure, in atm? Show the calculation and tick the correct one. (3)

(i) 35 atm (ii) 25 atm (iii) 45 atm (iv) 55 atm.

4. The half life of radium is 1590 years. What is the value of decay constant? (Show the calculation) (2)

5. The vapor pressure of pure liquid solvent a is 0.80 atm. When a nonvolatile substance B is added to the solvent, its vapor pressure drops to 0.60 atm. What is the mole fraction of component of B in the solution? Show the calculation and tick the correct one. (3)

(i) 1.25 (ii) 0.25 (iii) 0.75 (iv) 1.00

6. For the reaction at 298K, $2A + B \rightarrow C$, $\Delta H = 100 \text{ kCal}$, $\Delta S = 0.050 \text{ kcal/K}$. Assuming these values remain constant over the temperature range, at what temperature will the reaction become spontaneous? Show the calculation and tick the correct one. (3)

(i) 200 K (ii) 600 K (iii) 1000 K (iv) 2000 K

7. Calculate the normality of 7.88g of HNO_3 , per L, solution. (Formula weight of HNO_3 is 63.02 g/eq) Show the calculation and tick the correct one. (2)

(i) 0.15N (ii) 0.125N (iii) 0.0125N (iv) 0.015N

8. In a certain process, 678 J of heat is absorbed by a system while 294 J of work is done on the system. What is the change in internal energy for the process? Heat added to the system is considered as positive. Show the calculation and tick the correct one (2)

(i) 792 J (ii) 892 J (iii) 972 J (iv) 872 J

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(Please write each section on separate answer books)

Section -I

- Que.1 (a) Show that the volume ratio of two gases, before mixing and at the same temperature and pressure is equal to the pressure ratio after the two gases are mixed. (4)
- (b) Determine the density of H_2S gas at 27°C and 2.00 atm pressure. (M.W. of H_2S = 34.0 g/mol, Gas constant $R = 0.0821 \text{ L atm/K-Mol}$) (3)
- (c) Calculate the temperature at which 2.00 L samples of gas at 27°C would occupy 3.00L if its pressure were changed from 760 torr to 800 torr. (3)

OR

- Que.1 (a) Obtain the combined gas law from Charles law and Gay-Lussac's law. (4)
- (b) What volume of hydrogen will combine with 22 L chlorine to form hydrogen chloride? What volume of HCl will be formed? Assume the same temperature and pressure for all gases. (3)
- (c) A 1.00 L vessel containing 1.00g H_2 gas at 27°C is connected to a 2.00 L vessel containing 88.0 g CO_2 gas, also at 27°C . When the gases are completely mixed, what are the pressure and total pressure (in atm)? (M.W. of $\text{H}_2 = 2$ and $\text{CO}_2 = 44$) (3)

- Que.2 (a) Show that for an ideal gas enthalpy (H) is a function of temperature only and find relation between specific heat and constant pressure and volume for an ideal gas. (4)
- (b) For any PVT system show that change in entropy $dS = C_v dT/T$ for a process at constant volume. (3)
- (c) For the reaction at 298 K, $\text{A(g)} + \text{B(g)} \rightarrow \text{E(g)}$, $\Delta E = -3.00 \text{ kcal}$, $\Delta S = -10.0 \text{ kcal/K}$ calculate ΔG . $R = 1.987 \text{ cal/mol-K}$ (3)

Section -II

- Que. 3 (a) State second law of thermodynamics. (3)
- (b) Given the following information,
 $A + B \rightarrow C + D$: $\Delta H^0 = -10.0$ kcal, and $C + D \rightarrow E$: $\Delta H^0 = -10.0$ kcal
calculate ΔH^0 for (i) $A + B \rightarrow E$ (ii) $2C + 2D \rightarrow 2A + 2B$ (4)
- (c) If 1.0 kcal of heat is added to 1.2 L of oxygen in a cylinder at constant pressure of 1.0 atm the volume increases to 1.5 L. Calculate ΔE for the process. (3)
- Que. 4 (a) Distinguish between (i) the boiling point of a liquid and the normal boiling point of the liquid (ii) freezing point and freezing point depression. (4)
- (b) A solution containing 3.5 g of solute X in 50.0 g of water has a volume of 52.5 mL and a freezing point of -0.86°C . Calculate the molality, mole fraction and molarity of X. (3)
- (c) The vapor pressure of pure water at 25°C is 23.76 torr. The vapor pressure of a solution containing 5.4 g of a non-volatile substance in 90.0 g water is 23.33 torr. Compute the molecular weight of the solute. (3)
- Que.5 (a) In terms of reaction kinetics, explain why each of the following speeds up chemical reactions. (i) Catalyst (ii) increase in temperature and (iii) increase in concentration. (4)

OR

- (a) What is the time required for half the reactants to undergo second order reaction, when they are present in equal amounts. (4)
- (b) Deduce the rate law expression for conversion of H_2 and I_2 @ 400°C corresponds to,
(i) $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$ (one step) (ii) $\text{I}_2 \leftrightarrow 2\text{I}$, $2\text{I} + \text{H}_2 \rightarrow 2\text{HI}$ (two steps) (3)
- (c) Calculate the activation energy for a reaction which doubles in rate when the temp is raised from 18 to 28°C . (3)
